

Collected Mathematical Papers, Vol. XII

PDF pages 326–335 (printed pp. 306–315)

Papers 837 (conclusion), 838, 839, 840 (start)

Arthur Cayley

306 ON THE SO-CALLED D’ALEMBERT–CARNOT GEOMETRICAL PARADOX. [837

the two points m thus determined are on one and the same line $t'Kt$; but there is no *a priori* reason why the positive senses should be the same, and they are in fact opposite to each other, in the two cases respectively; in the one case we measure off the distance ρ_1 in the sense Kt , in the other case the distance ρ_2 in the sense $+Kt'$; that is, we in fact measure off the positive distances $+\rho_1$ and $+\rho_2$, in one and the same sense Kt ; thus obtaining for the point m one or the other extremity of a determinate secant through K .

The best illustration is I think in the elementary problem of finding the perpendicular distance of a given line from the origin. Let $Ax + By + C = 0$ be the equation of the given line: and first let a line be drawn in a determinate sense, say at the inclination θ to the positive part of the axis of x , to meet the given line. Taking r for the distance from the origin of the point of intersection, we have, for the coordinates of the point of intersection, $x, y = r \cos \theta, r \sin \theta$; and thence

$$r(A \cos \theta + B \sin \theta) + C = 0,$$

that is,

$$r = \frac{-C}{A \cos \theta + B \sin \theta},$$

a perfectly determinate value. But the perpendicular on the given line may be considered as drawn in at pleasure one or the other of two opposite senses; that is, we have

$$\cos \theta, \sin \theta = \frac{A}{\sqrt{A^2 + B^2}}, \frac{B}{\sqrt{A^2 + B^2}},$$

or else

$$= \frac{-A}{\sqrt{A^2 + B^2}}, \frac{-B}{\sqrt{A^2 + B^2}};$$

and thence $r = \frac{-C}{\sqrt{A^2 + B^2}}$, or else $r = \frac{+C}{\sqrt{A^2 + B^2}}$; that is, the perpendicular distance is $= \frac{\pm C}{\sqrt{A^2 + B^2}}$, with the essentially indeterminate sign $\pm$, because the distance may be considered as drawn from the origin in one or the other of the two opposite senses.

838.

ON THE TWISTED CUBICS UPON A QUADRIC SURFACE.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 129–132.]

A CUBIC (twisted cubic) on a quadric surface meets each generator of the one kind twice, and each generator of the other kind once (Salmon, *Solid Geometry*, Ed. 4, p. 301). There are thus on the surface two kinds of cubics, viz. distinguishing for convenience the generating lines as generators and directors, a cubic may meet each generator twice and each director once; or it may meet each generator once and each director twice. And two cubic curves are accordingly of the same kind or of different kinds.

Consider for example the quadric surface $xw - yz = 0$, and the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

If we call the line $(x = ky, kw = z)$ a generator, and the line $(x = kz, kw = y)$ a director, for the intersections with a generator we have $1 = k\phi, k\phi^2 = \phi^3$; equations with the common root $k\phi = 1$, or there is a single intersection; for the intersections with a director, $1 = k\phi^2, k\phi = \phi^3$, equations with the common roots $k\phi^2 = 1$, or there are two intersections.

Consider on the same quadric surface the cubic

$$x : y : z : w = \theta - \alpha.\theta - \beta.\theta - \gamma : \theta - \alpha.\theta - \epsilon.\theta - \zeta : \theta - \delta.\theta - \beta.\theta - \gamma : \theta - \delta.\theta - \epsilon.\theta - \zeta;$$

or as, for shortness, I write these,

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta,$$

viz. α is written to denote $\theta - \alpha$, and so in other cases; for the intersections with a generator, we have $\alpha\beta\gamma = k\alpha\epsilon\zeta, k\delta\epsilon\zeta = \delta\beta\gamma$, equations having the two common roots

$\beta\gamma = k\epsilon\zeta$; or there are two intersections with the generator. And in like manner there is one intersection with the director. The cubic

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$$

is thus a cubic of different kind from

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

And in the same manner it appears that the cubic

$$x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$$

is of the same kind with

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

The two cubics of different kinds intersect in 5 points; the two cubics of the same kind in only 4 points. In fact, for the intersections with the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3,$$

we have $xz - y^2 = 0$, $yw - z^2 = 0$, and substituting herein the θ -values,

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta,$$

we find

$$\alpha\delta\beta^2\gamma^2 - \alpha^2\epsilon^2\zeta^2 = 0, \quad \alpha\epsilon\zeta\delta\epsilon\zeta - \delta^2\beta^2\gamma^2 = 0,$$

equations with the common five roots $\delta\beta^2\gamma^2 - \alpha\epsilon^2\zeta^2 = 0$; whereas for the θ -values $x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$, we have $\alpha\delta\beta^2\gamma^2 - \alpha\delta\beta\gamma\epsilon\zeta = 0$, $\delta\beta\gamma\epsilon\zeta - \alpha^2\epsilon^2\zeta^2 = 0$, equations with the common four roots $\alpha\epsilon\zeta - \delta\beta\gamma = 0$.

I remark in passing that the equations just obtained have each of them a root $\theta = \infty$, but this would have been avoided if the θ -equations had been taken in the slightly altered form

$$x : y : z : w = \alpha\beta\gamma : b\alpha\epsilon\zeta : c\delta\beta\gamma : bc\delta\epsilon\zeta,$$

and

$$x : y : z : w = \alpha\beta\gamma : c\delta\beta\gamma : b\alpha\epsilon\zeta : bc\delta\epsilon\zeta,$$

b, c being arbitrary constants; and the special value is thus of no importance.

The two cubics intersecting in 5 points constitute the complete intersection of a quadric surface and a cubic surface; and conversely when a quadric surface and a cubic surface intersect in two cubics, then the cubics must have 5 intersections, and are thus by what precedes cubics of different kinds in regard to the quadric surface. In fact, considering two cubics meeting in 5 points, we can, through these 5 points, through 2 points assumed at pleasure on the first cubic, and through 2 points assumed at pleasure on the second cubic, draw a determinate quadric surface: this meets each of the two cubics in $5 + 2$ points, and consequently it entirely contains the cubic; viz. we have through the two cubics a determinate quadric surface. Again, through the 5 points, through 5 points at pleasure on the first cubic and through 5 points at pleasure on the second cubic, we may draw a cubic surface; this meets each cubic

in 5 + 5 points, and therefore it entirely contains the cubic; we have thus a cubic surface through the two cubics. The cubic surface has been subjected only to $5 + 5 + 5 = 15$ conditions, and the equation thus contains homogeneously $20 - 15 = 5$ constants; viz. if $\Theta = 0$ be the quadric surface through the two cubics the general form is $(\alpha x + \beta y + \gamma z + \delta w)\Theta + \lambda U = 0$; disregarding the term in Θ , we have thus, in fact, a determinate cubic $U = 0$.

Taking as above the first cubic as given by the equations

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3,$$

and the second cubic as given by the equations

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta,$$

the equation of the cubic surface which contains the two cubics will be of the form

$$(Ax + By + Cz + Dw)(yw - z^2) + (A''x + B''y + C''z + D''w)(xz - y^2) = 0.$$

Substituting herein the θ -values and omitting the common factor $\alpha\epsilon\zeta^2 - \delta\beta^2\gamma^2$, we have

$$\delta(A\alpha\beta\gamma + B\alpha\epsilon\zeta + C\delta\beta\gamma + D\delta\epsilon\zeta) - \alpha(A''\alpha\beta\gamma + B''\alpha\epsilon\zeta + C''\delta\beta\gamma + D''\delta\epsilon\zeta) = 0,$$

which is of the fourth order in θ . We may assume $C'' = 0$ and $D'' = 0$; in fact, the terms in C'' and D'' might be got rid of by the substitutions

$$z(xz - y^2) = y(xw - yz) - x(yw - z^2),$$

$$w(xz - y^2) = z(xw - yz) - y(yw - z^2);$$

we then have 6 coefficients A, B, C, D , and A'', B'' , equating to zero the terms in $\theta^0, \theta^1, \theta^2, \theta^3, \theta^4$, the ratios of these are determined by the five equations: and we have thus the required cubic surface.

Starting from either cubic considered as a curve on the given quadric surface; if through the centre we describe a cubic surface, this meets the quadric surface in a second cubic and the two cubics intersect each other in 5 points. In particular, if the one cubic is $x : y : z : w = 1 : \phi : \phi^2 : \phi^3$, it appears by what precedes that the other cubic may be taken to be

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta,$$

intersecting the former in the 5 points given by the equation $\delta\beta^2\gamma^2 - \alpha\epsilon^2\zeta^2 = 0$. The five points are of course points of contact of the quadric surface and the cubic surface.

A very simple instance is the following. The quadric surface $xw - yz = 0$ and the cubic surface

$$y(xz - y^2) + z(yw - z^2) = 0$$

intersect in the two cubics

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3 \quad \text{and} \quad x : y : z : w = 1 : \theta' : \theta'' : \theta'''.$$

The two cubics meet in the 5 points

$$\theta = (0, \infty, 1, \omega, \omega^2); \quad \phi = (0, \infty, 1, \omega^2, \omega),$$

or, what is the same thing, in the 5 points

$$(x, y, z, w) = (1, 0, 0, 0), (0, 0, 0, 1), (1, 1, 1, 1), (1, \omega, \omega^2, 1), (1, \omega^2, \omega, 1),$$

where ω is an imaginary cube root of unity.

I was anxious to work out this result not for its own sake, but as an illustration of a point which first arises as to octics: a unicursal octic curve is not in general situate on any surface lower than a quartic surface; if on the octic we take at pleasure 33 points, we may through these draw two quartic surfaces $U = 0$, $V = 0$, each of which will entirely contain the curve: the two surfaces meet in a second octic also unicursal, and the two curves intersect in 34 points, which are points of contact of the two quartic surfaces. We cannot, by adjoining to the given octic any curve of an inferior order to its own, obtain a complete intersection of two surfaces: the most simple case is when we adjoin to the given octic as above another unicursal octic, and thus obtain a complete intersection of two quartic surfaces. It is to be observed that the second curve is determined completely and uniquely by the given curve: considering the former curve as given by the expressions of the coordinates in terms of a parameter ϕ , the determination of the like expressions for the latter curve would appear to be a problem of very great difficulty.

839.

ON THE MATRICAL EQUATION $qQ - Qq' = 0$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 176–178.]

I CONSIDER the matrical equation $qQ - Qq' = 0$, where q, q' are given matrices

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}, \quad \begin{vmatrix} a' & b' \\ c' & d' \end{vmatrix}, \quad \text{and} \quad Q = \begin{vmatrix} A & B \\ C & D \end{vmatrix}$$

is a matrix which has to be determined. As remarked in the paper “On the Quaternion Equation $qQ - Qq' = 0$,” *Messenger*, t. XIV. (1885), pp. 108–112, [836] the question for matrices is equivalent to that for quaternions: for a matrix $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ may be regarded as a quaternion

$$\frac{1}{2}a(1 - \lambda i) + \frac{1}{2}b(j + \lambda k) + \frac{1}{2}c(-j - \lambda k) + \frac{1}{2}d(1 + \lambda i),$$

or (omitting the factor $\frac{1}{2}$) as the quaternion

$$(a + d) - \lambda(a - d)i + (b - c)j - \lambda(b + c)k,$$

where $\lambda, = \sqrt{-1}$, is the imaginary of ordinary algebra. Hence considering q, q' as denoting the quaternions

$$(a + d) - \lambda(a - d)i + (b - c)j - \lambda(b + c)k,$$

$$(a' + d') - \lambda(a' - d')i + (b' - c')j - \lambda(b' + c')k,$$

we can, if a certain condition is satisfied, find a quaternion Q such that $qQ - Qq' = 0$; say this is $Q = W + iX + jY + kZ$; putting this

$$= \frac{1}{2}\{(A + D) - \lambda(A - D)i + (B - C)j - \lambda(B + C)k\},$$

we find

$$Q = \begin{vmatrix} A & B \\ C & D \end{vmatrix} = \begin{vmatrix} W + \lambda X & Y + \lambda Z \\ -Y + \lambda Z & W - \lambda X \end{vmatrix},$$

for the required matrix Q ; this being an indeterminate matrix, such that $AD - BC = 0$.

But it is better to solve directly the matrical equation

$$\begin{vmatrix} a, & b \\ c, & d \end{vmatrix} \begin{vmatrix} A, & B \\ C, & D \end{vmatrix} - \begin{vmatrix} A, & B \\ C, & D \end{vmatrix} \begin{vmatrix} a', & b' \\ c', & d' \end{vmatrix} = 0,$$

viz.

$$\begin{vmatrix} (a,b)(A,C), & (a,b)(B,D) \\ (c,d) \text{ " }, & (c,d) \text{ " } \end{vmatrix} - \begin{vmatrix} (a',c')(A,B), & (b',d')(A,B) \\ \text{ " }, & \text{ " } \end{vmatrix} = 0,$$

that is,

$$\begin{aligned} Aa + Cb - (Aa' + Bc') &= 0, \\ Ba + Db - (Ab' + Bd') &= 0, \\ Ac + Cd - (Ca' + Dc') &= 0, \\ Bc + Dd - (Cb' + Dd') &= 0, \end{aligned}$$

or, what is the same thing,

$$\begin{vmatrix} a - a', & -c', & 0, & b \\ -b', & a - d', & 0, & b \\ c, & 0, & d - a', & -c' \\ 0, & c, & -b', & d - d' \end{vmatrix} (A, B, C, D) = 0,$$

viz. we have (A, B, C, D) connected by these four linear equations: viz. the necessary condition is that the determinant formed out of the matrix which here presents itself shall be $= 0$.

After some reductions, and putting for shortness

$$\nabla = ad - bc, \quad \nabla' = a'd' - b'c',$$

this is found to be

$$(\nabla - \nabla')^2 + [\nabla(a' + d') - \nabla'(a + d)](a' + d' - a - d) = 0;$$

which is the condition for the existence of a solution. This condition being satisfied, the four equations will be equivalent to two independent equations, which serve to determine A, B, C, D ; and, assuming the absolute value of A , we find

$$\begin{aligned} A &= -(a' + d' - a - d), \\ c'B &= \nabla - \nabla' + a'(a' + d' - a - d), \\ b'C &= \nabla - \nabla' + a(a' + d' - a - d), \\ b'c'D &= (\nabla - aa')\nabla - (d - a')\nabla' - aa'(a' + d' - a - d), \end{aligned}$$

values which give

$$-bc(AD - BC) = (\nabla - \nabla')^2 + [\nabla(a' + d') - \nabla'(a + d)](a' + d' - a - d) = 0,$$

viz. in the case when the given matrices satisfy the above-mentioned condition, the components A, B, C, D of the required matrix have determinate values which are such that $AD - BC = 0$.

839]

ON THE MATRICAL EQUATION $qQ - Qq' = 0$.

313

If we have $\nabla - \nabla' = 0$, that is, $ad - bc = a'd' - b'c'$, then the condition becomes $a + d = a' + d'$; and these two conditions being satisfied, the four equations reduce themselves to two independent equations, and the ratios $A : B : C : D$ have no longer determinate values; the linear equations may in fact be written

$$\begin{vmatrix} 0, & b, & -c', & a - a' \\ -b', & 0, & -a + d', & b \\ c, & a' - d, & 0, & -c' \\ d - d', & -b', & c, & 0 \end{vmatrix} \cdot (D, C, B, A) = 0,$$

which are of the form

$$\begin{vmatrix} 0, & h, & -g, & a \\ -h, & 0, & f, & b \\ g, & -f, & 0, & c \\ -a, & -b, & -c, & 0 \end{vmatrix} \cdot (D, C, B, A) = 0,$$

where the coefficients a, b, c, f, g, h are such that $af + bg + ch = 0$; and the four equations are thus equivalent to two independent equations. To obtain a symmetrical solution, assume a relation

$$D\xi + C\eta + B\zeta + A\omega = 0,$$

with arbitrary coefficients ξ, η, ζ, ω ; we then find

$$\begin{aligned} D &= \cdot - c\eta - b\zeta + f\omega, \\ C &= -c\xi \cdot + a\zeta + g\omega, \\ B &= -b\xi - a\eta \cdot + h\omega, \\ A &= f\xi - g\eta - h\zeta \cdot, \end{aligned}$$

viz. the complete theorem is that, if the matrices

$$q = \begin{vmatrix} a, & b \\ c, & d \end{vmatrix}, \quad q' = \begin{vmatrix} a', & b' \\ c', & d' \end{vmatrix}$$

are such that $ad - bc = a'd' - b'c'$, $a + d = a' + d'$, then writing

$$\begin{aligned} \mathfrak{a} &= a - a', & \mathfrak{f} &= -d + d', & \mathfrak{f} &= a + d = a' + d', \\ \mathfrak{b} &= b', & \mathfrak{g} &= -c', \\ \mathfrak{c} &= -c, & \mathfrak{h} &= b, \end{aligned}$$

values which satisfy

$$\mathfrak{a}\mathfrak{f} + \mathfrak{b}\mathfrak{g} + \mathfrak{c}\mathfrak{h} = 0,$$

and taking ξ, η, ζ, ω arbitrary, we have for the matrix Q , which is such that $Qq' - Qq = 0$, the expression

$$Q = \begin{vmatrix} -\mathfrak{f}\xi - \mathfrak{g}\eta - \mathfrak{h}\zeta, & \mathfrak{b}\xi - \mathfrak{a}\eta \cdot + \mathfrak{h}\omega \\ -\mathfrak{c}\xi \cdot + \mathfrak{a}\zeta + \mathfrak{g}\omega, & \cdot - c\eta - \mathfrak{b}\zeta + \mathfrak{f}\omega \end{vmatrix},$$

depending really on one arbitrary parameter, viz. we may without loss of generality put any two of the coefficients ξ, η, ζ, ω equal to 0.

840.

ON MASCHERONI'S GEOMETRY OF THE COMPASS.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 179–181.]

I HAVE not seen the original of Mascheroni's work, *La Geometria del Compasso*, 8°, Pavia (1797), but only the French translation, *Géométrie du Compas*, par *L. Mascheroni*, traduite de l'Italien par *A. M. Carette*, 2^e éd. 8° Paris, 1828 (Author's Preface, pp. 5–24, and pp. 25–328, with 14 plates containing 108 figures). The title expresses the notion of the work: a straight line is given by means of two points thereof, but is not allowed to be actually drawn; and the problem is, with the compass alone to perform all the constructions of Geometry. Observe that, for the purpose of demonstration, any lines may be imagined to be drawn, and such lines are in fact shown as dotted lines in the figures; but this does not in any wise interfere with the fundamental postulate that the constructions are to be performed with the compass only.

Assuming, then, that a line is in every case given by means of two points thereof, the leading questions are those considered Book vii. on “the intersections of straight lines with circular arcs and with each other,” viz. they are (1) to find the intersections of a given line and circle; (2) to find the intersection of two given lines. But in the first problem it is necessary to consider two cases; (A) the general case, where the line does not pass through the centre of the circle, and (B) the particular, but actually more difficult, case, where the line passes through the centre of the circle. It is assumed that the centre of the circle is given: if it is not given, it can be found as afterwards mentioned.

It will be convenient to establish the definition of *counter-points* in regard to a given line: two points, which are such that the line joining them is bisected at right angles by the given line, are said to be counter-points in regard to that line.

(1), A. Consider a line given by means of two points P , Q ; and a given circle, centre C . With centre P and radius PC describe a circle, and with centre Q and

radius QC a circle; these meet in C and in a second point D which will be the counter-point of C in regard to the line PQ ; hence with centre D and radius = that of given circle, describing a circle, this meets the given circle in two points which are the intersections of the given circle by the line PQ .

The construction fails if the line PQ passes through the centre of the given circle, for then the two auxiliary circles touch at C , or we have D coincident with C . We have therefore considered this case; Q may be taken to be the centre C of the circle; so that we have:

(1), B. The problem is, given a point P , and a circle, centre C : to find the intersections of the circle by the line CP . With centre P , and an arbitrary radius, describe a circle meeting the given circle in points A, B (which are of course counter-points in regard to line CP); the circle is divided into two arcs AB and $360^\circ - AB$, and the problem is to bisect each of these two arcs; for the points of bisection are obviously the required intersections of CP with the circle.

Hence we have:

(1), C. To bisect a given arc AB of a circle, centre C (fig. 1). With centre A and radius AC describe a circle; and with centre B , and equal radius BC describe a circle; and on these circles respectively, find the points D, E such that $CD = CE = AB$.

[Figure 1: an arc AB of a circle with centre C ; two circles centred at A and B both passing through C ; points D, E marked on these auxiliary circles so that $CD = CE = AB$; auxiliary point F above is the second intersection of the two larger circles of radii EA, DB .]

With centre E and radius EA , describe a circle; and with centre D , and equal radius DB , describe a circle; and let these two circles meet in F ; then with centre E (or D) and radius GF describe a circle: this will meet the given circle in the required middle point G of the arc AB .

The proof depends on the theorem that in a rhombus the sum of the squares of the diagonals is = sum of the squares of the four sides. We have a rhombus $ACEB$, and therefore

$$(AE)^2 + (BC)^2 = 2(AB)^2 + 2(AC)^2,$$

that is,

$$(AE)^2 = 2(AB)^2 + \text{rad.}^2.$$

But by construction, $EF^2 = (AE)^2$; therefore

$$EF^2 = 2(AB)^2 + \text{rad.}^2 = (CE)^2 + \text{rad.}^2.$$

Hence in the right-angled triangle FCE , we have

$$(CF)^2 = (EF)^2 - CE^2 = (CE)^2 + \text{rad.}^2;$$

[Paper 840 continues on the following pages, which lie outside the present chunk.]